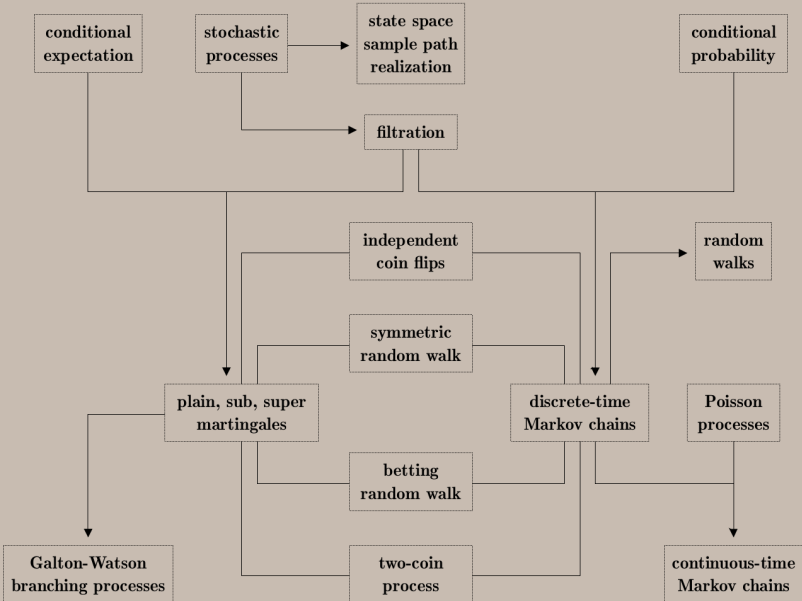
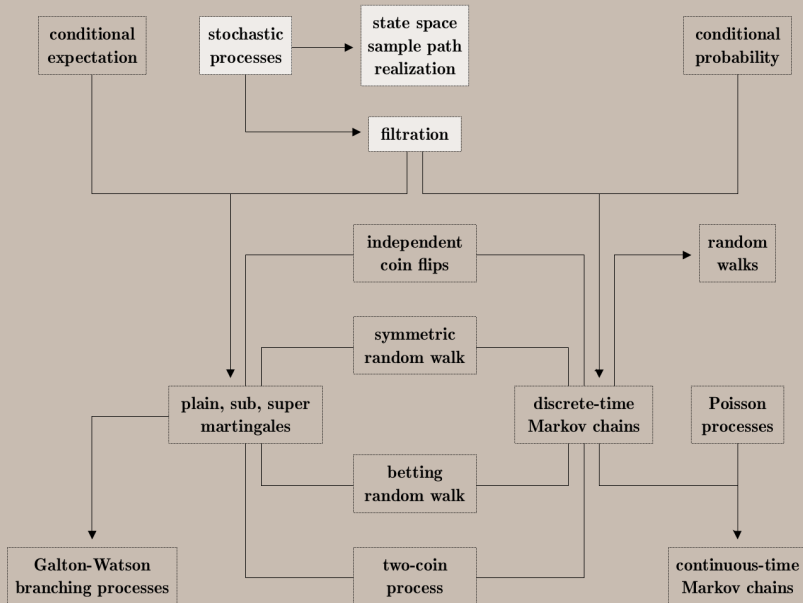




Paul Lévy



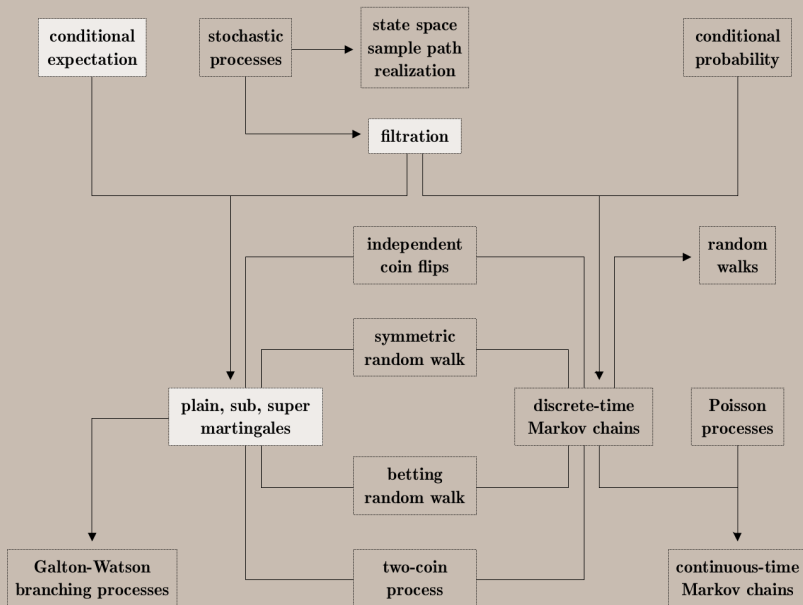
Andrey Markov



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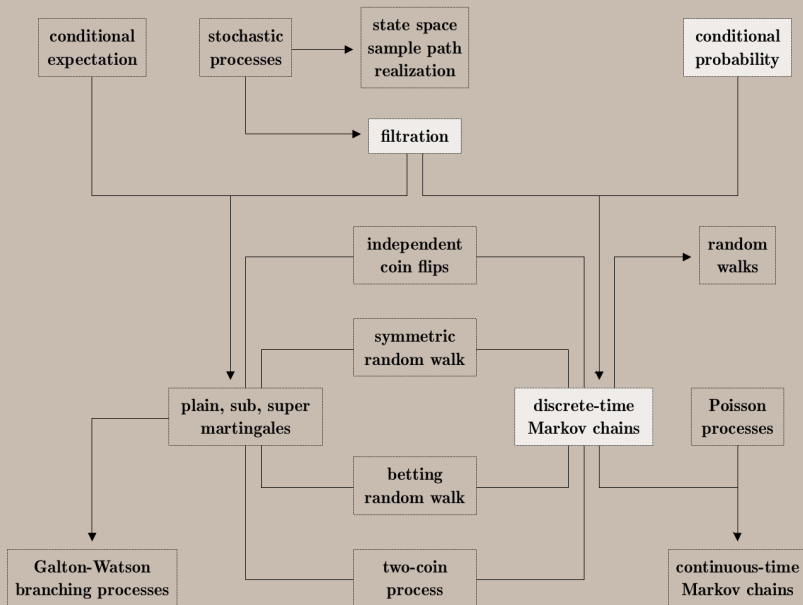
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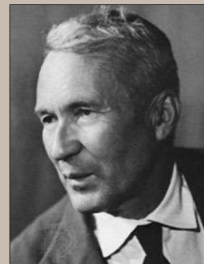
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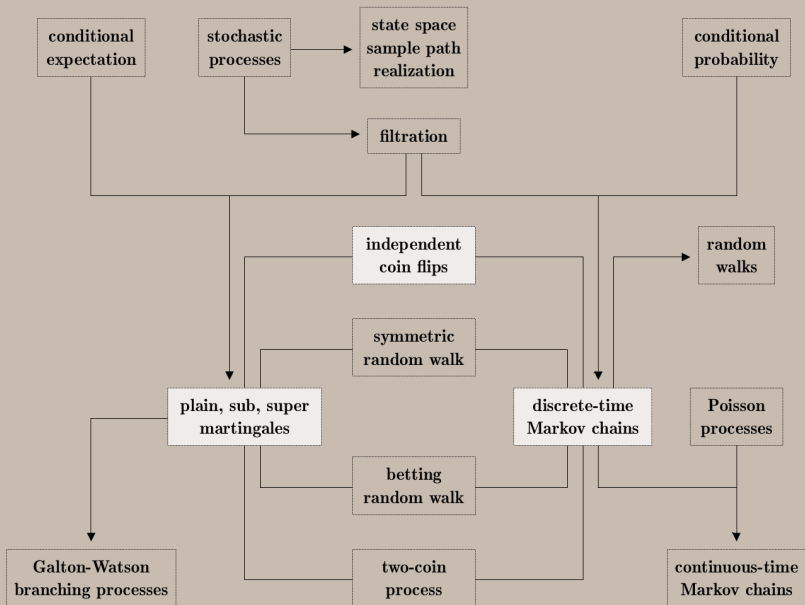
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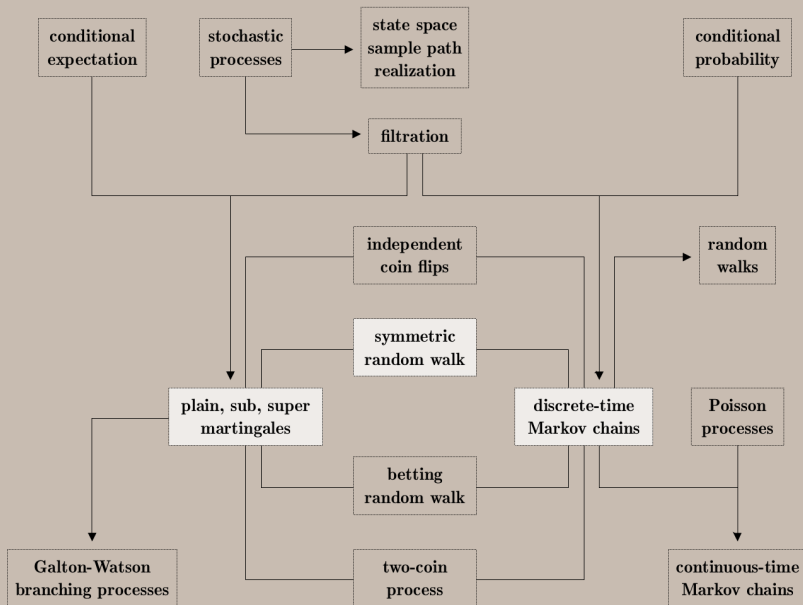
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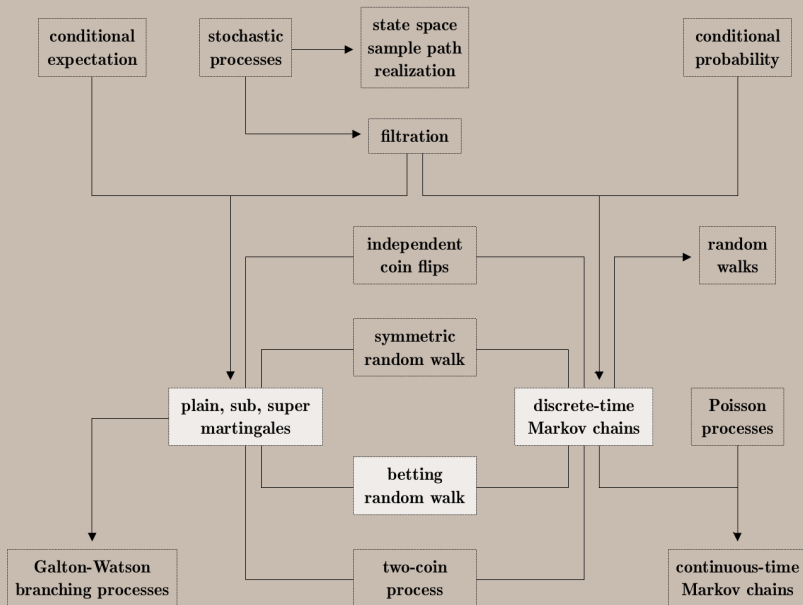
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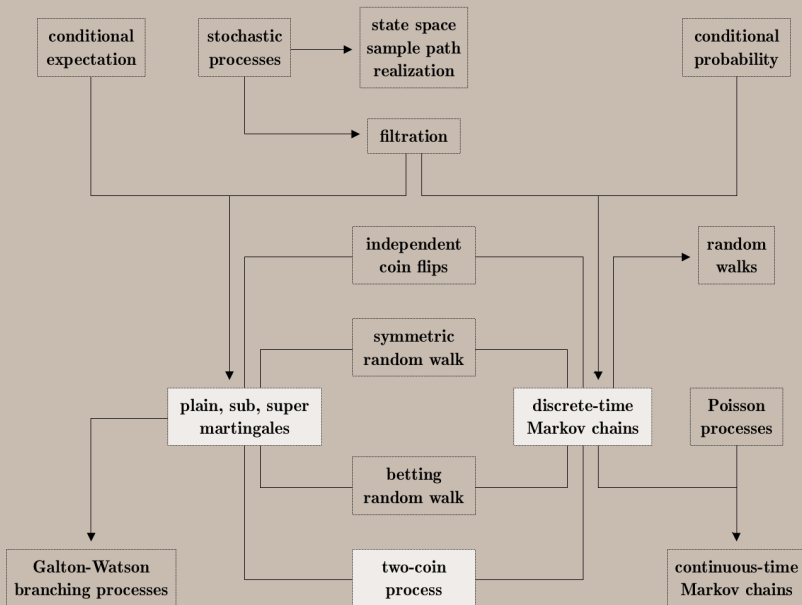
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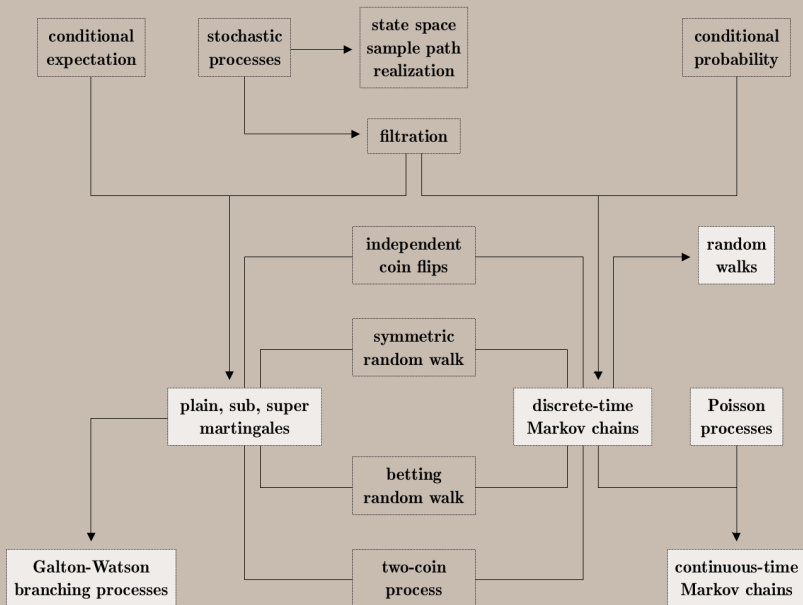
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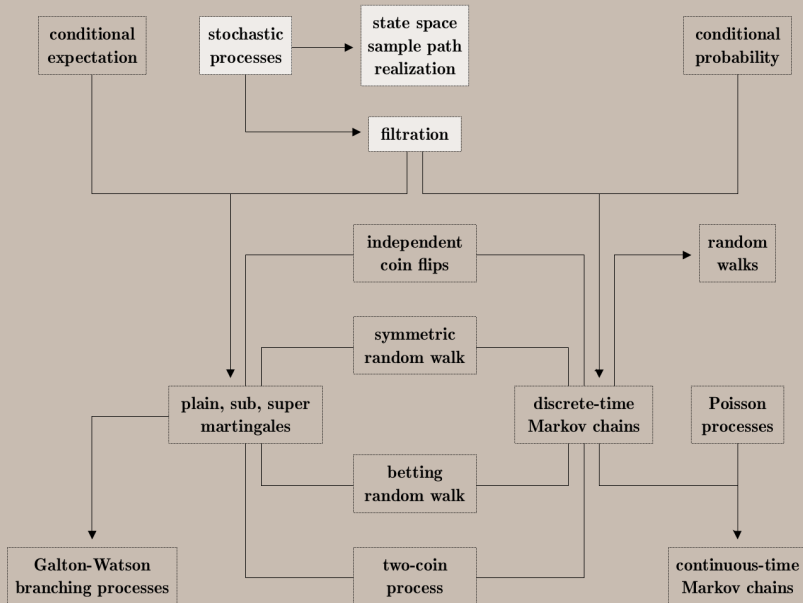
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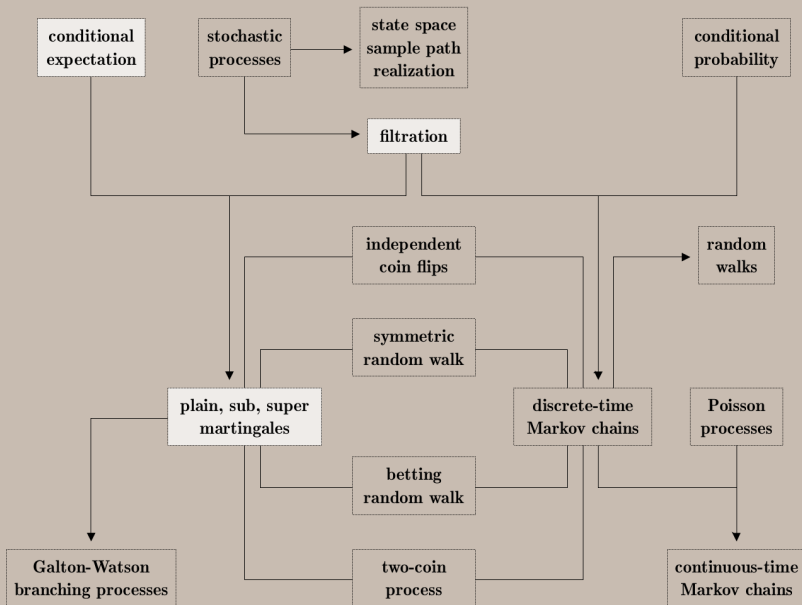
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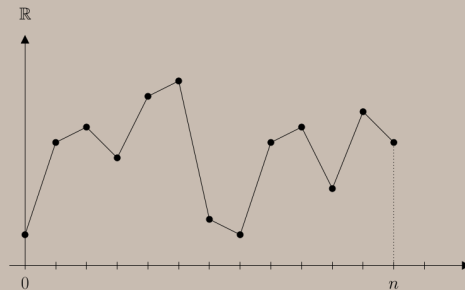
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Definition. **Martingale** w.r.t. a filtration $\mathcal{G}_n$:

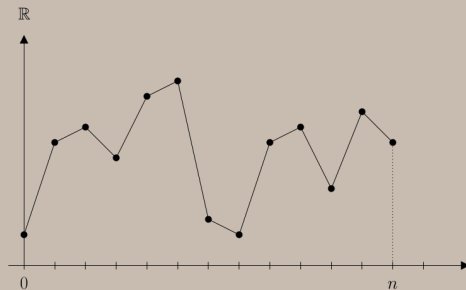
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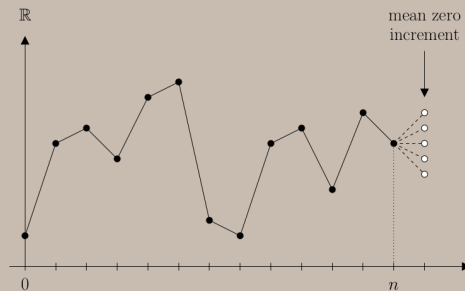
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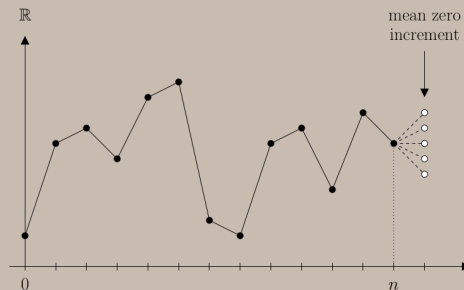
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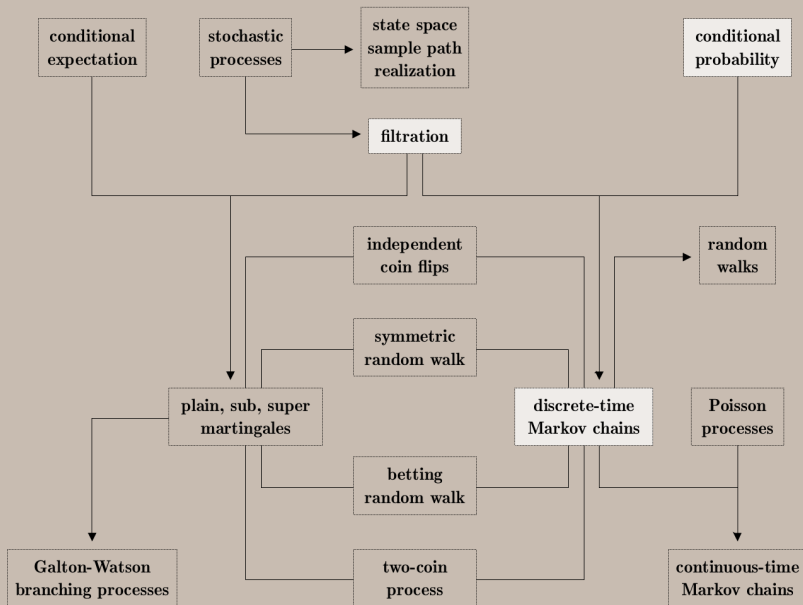
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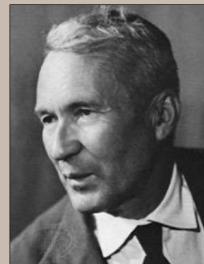
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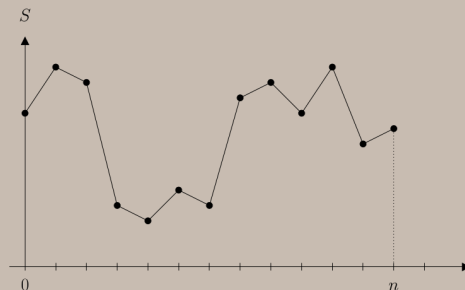
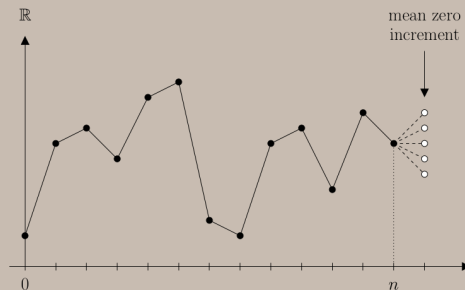
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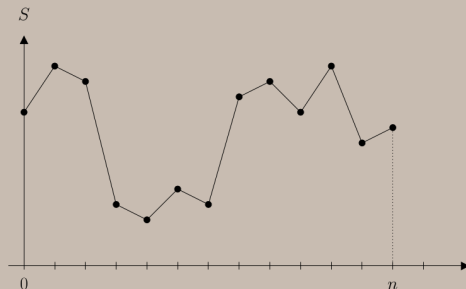
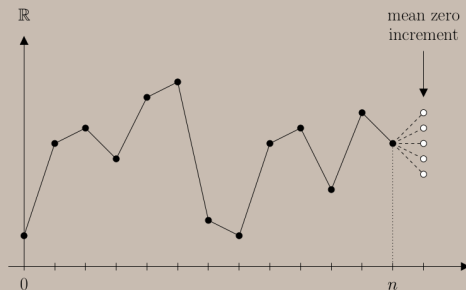
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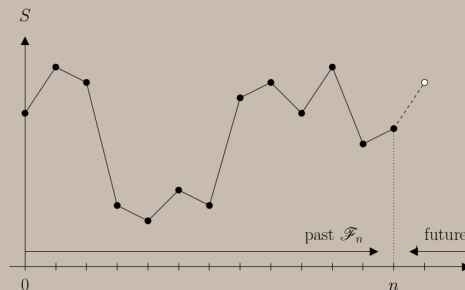
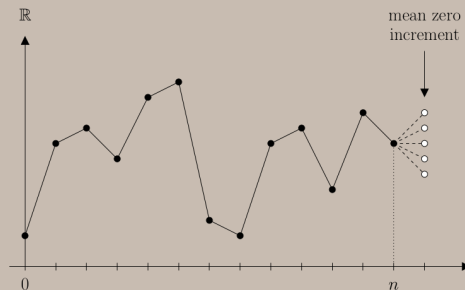
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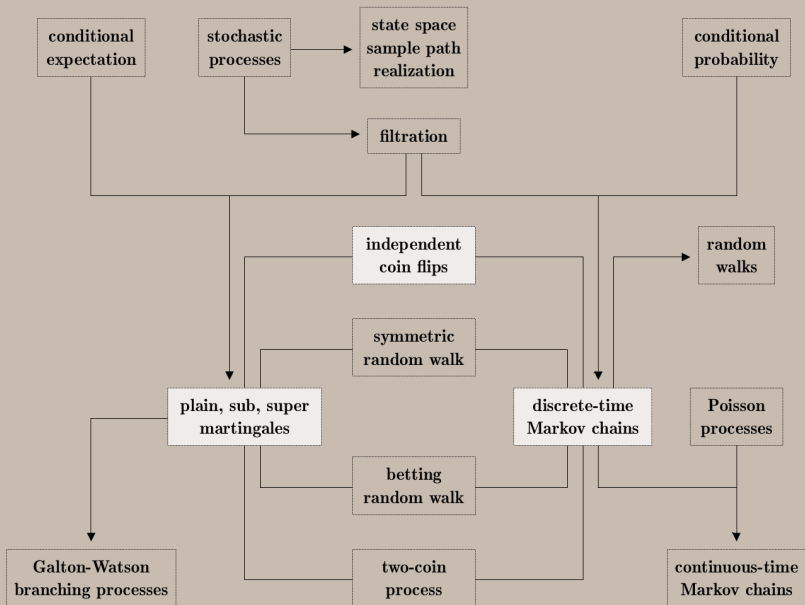
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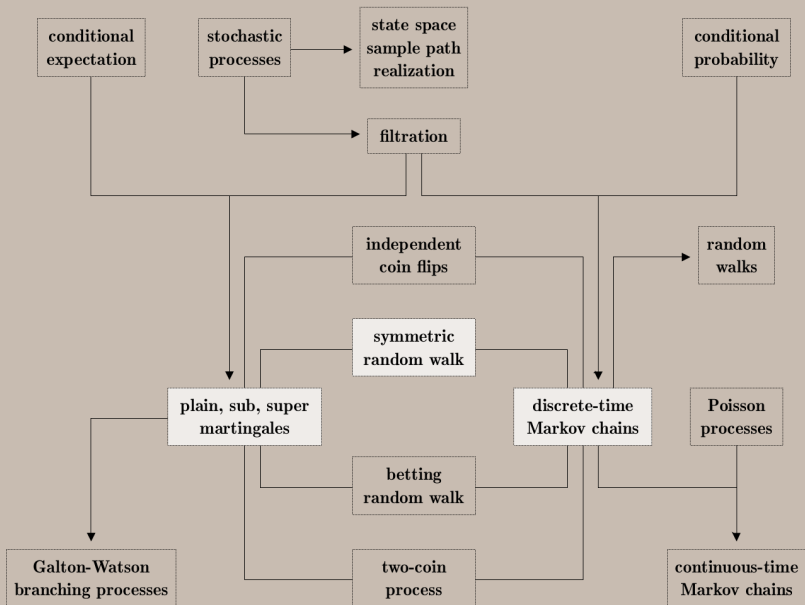
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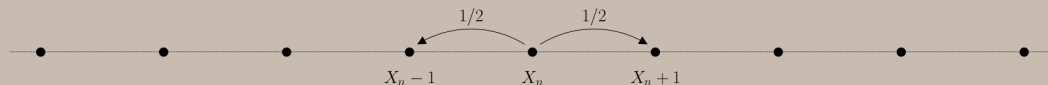


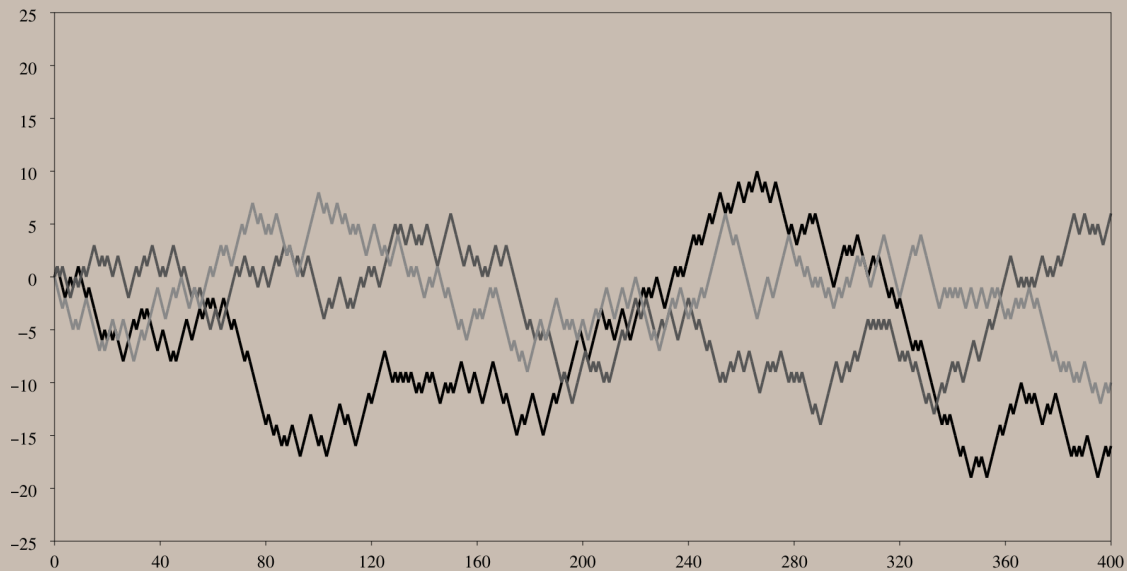
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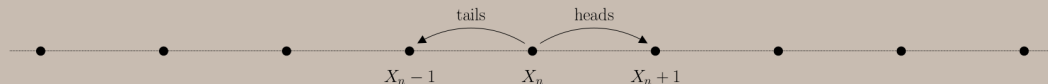


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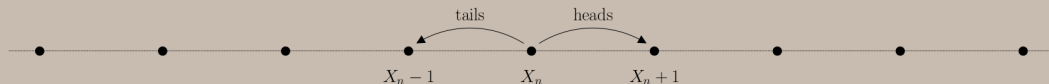


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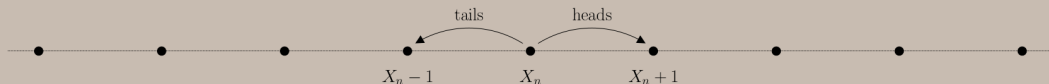


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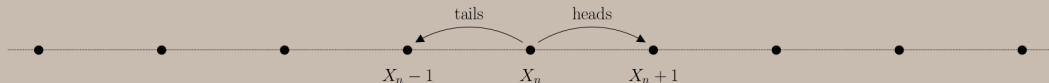


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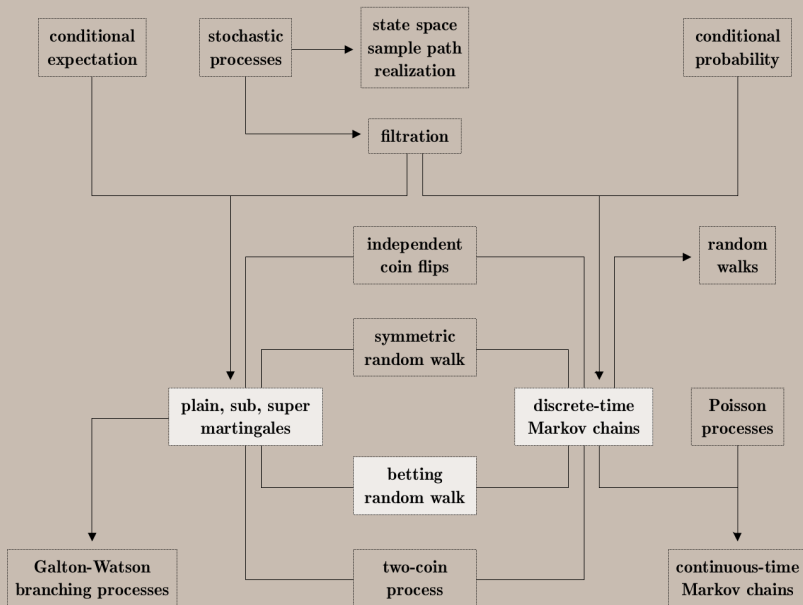
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Paul Lévy



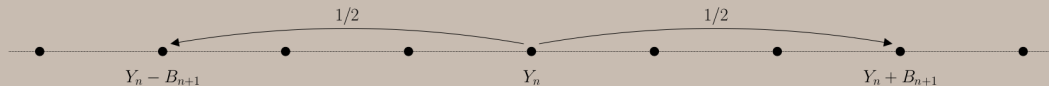
Andrey Markov

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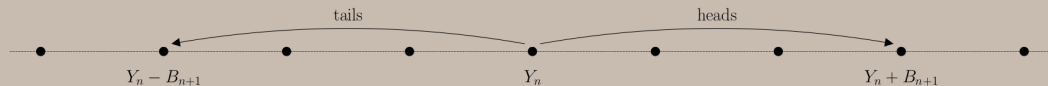


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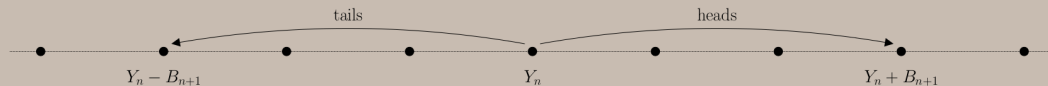


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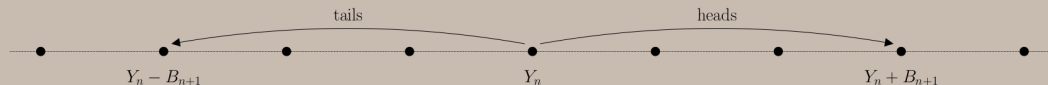
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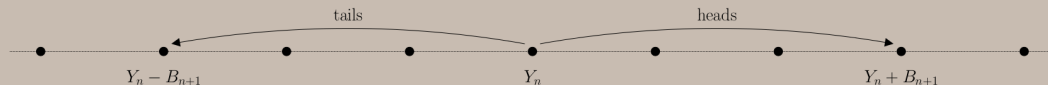
$$E(Y_{n+1} \mid \mathcal{F}_n) = E(Y_n + B_{n+1}W_{n+1} \mid \mathcal{F}_n) \stackrel{\text{IND}}{=} Y_n + B_{n+1} E(W_{n+1}) = Y_n \implies \text{martingale.}$$

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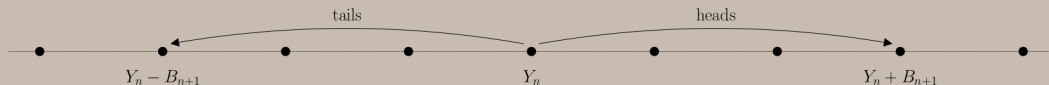


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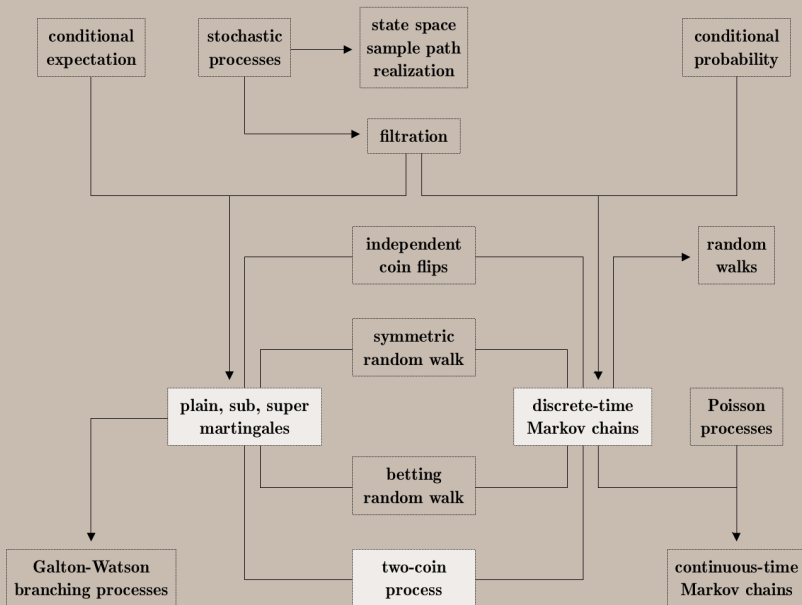
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$$\begin{aligned} P(Y_{n+1} = Y_n + a \mid \mathcal{F}_n) &= P(W_{n+1} = 1, B_{n+1} = a \mid \mathcal{F}_n) = E(\mathbf{1}\{W_{n+1} = 1\} \mathbf{1}\{B_{n+1} = a\} \mid \mathcal{F}_n) \\ &\stackrel{\text{IND}}{=} P(W_{n+1} = 1) \mathbf{1}\{B_{n+1} = a\} = \frac{1}{2} \mathbf{1}\{W_n = 1\} = \frac{1}{2} \mathbf{1}\{Y_n > Y_{n-1}\} \end{aligned}$$

$\implies$ **not a Markov chain.**



Paul Lévy



Andrey Markov

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Two-coin process: Assume that coin 1 comes up heads with probability one-third, and coin 2 comes up heads with probability two-thirds. Flip coin 1 twice, then flip

coin 1 when the previous two outcomes match,

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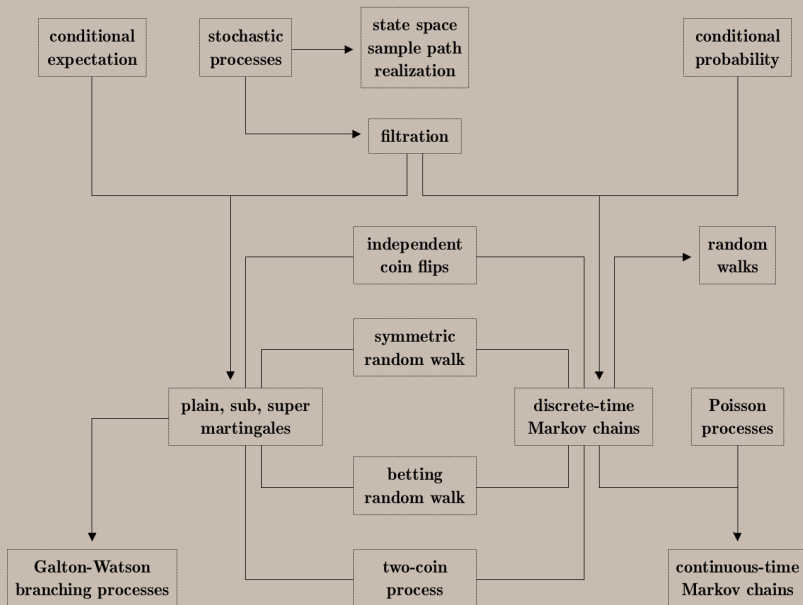
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